Weakly Supervised Deep Segmentation of Immunofluorescence Modalities Via Translation and Mask Fusion

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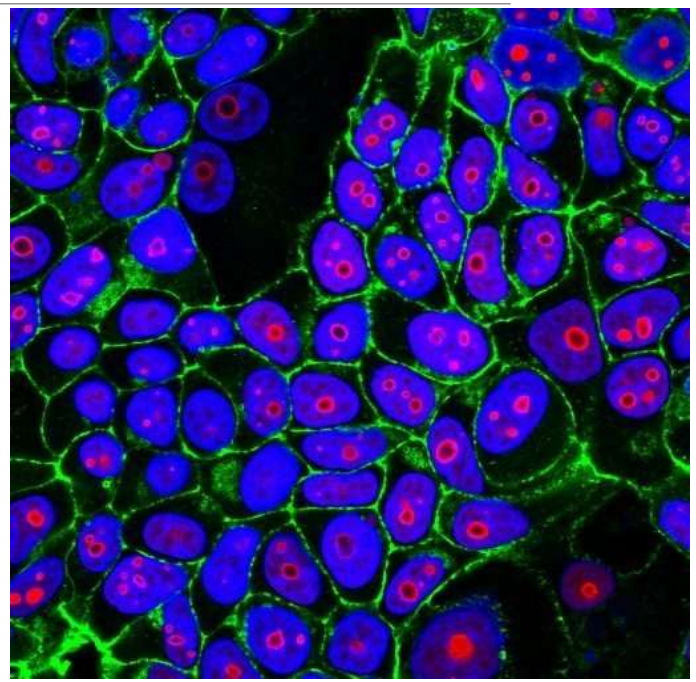
What is the project about?

1. It’s a weakly supervised, explainable deep learning framework for translating IHC images into fluorescence modalities and producing accurate cell-level segmentation to assist in disease diagnosis.
2. Immunofluorescence is a lab technique used to find specific proteins in cells or tissues. It uses special dyes that glow under a microscope, helping scientists see where certain proteins are located.
3. In immunofluorescence, antibodies linked to fluorescent dyes are added to a tissue sample. These antibodies stick only to the protein they match, and the dye makes that protein visible when viewed under a fluorescence microscope.



Objective

❖**Virtual Immunofluorescence Synthesis**

- Learn a mapping from IHC images to multiple fluorescence modalities (Hematoxylin, DAPI, LAP2, Ki67).

❖**Accurate Nuclei Segmentation**

- Perform cell-level (nuclei vs. background) segmentation using deep learning.
- Treat segmentation as a binary classification problem with robust overlap-based losses (Dice, IoU, AJI).

❖**Weakly Supervised Learning (WSL)**

- Demonstrate that weak supervision can still achieve competitive segmentation and translation performance.
- Reduce reliance on fully annotated pixel-level labels, which are costly and scarce in medical imaging.

❖**Comparative Evaluation of Advanced Architectures**

- Systematically compare multiple state-of-the-art architectures and attention blocks.

Related Works

- DeepLIIF: Deep Learning-Inferred Multiplex Immunofluorescence for IHC Image Quantification

Work Done: The researchers used AI to turn regular tissue images into detailed fluorescent images. They trained a model called DeepLIIF to identify cancer markers like Ki67, CD3, and BCL2 by analyzing and classifying cells automatically.

- Autofluorescence Virtual Staining System for H&E Histology and Multiplex Immunofluorescence in Lung Cancer.

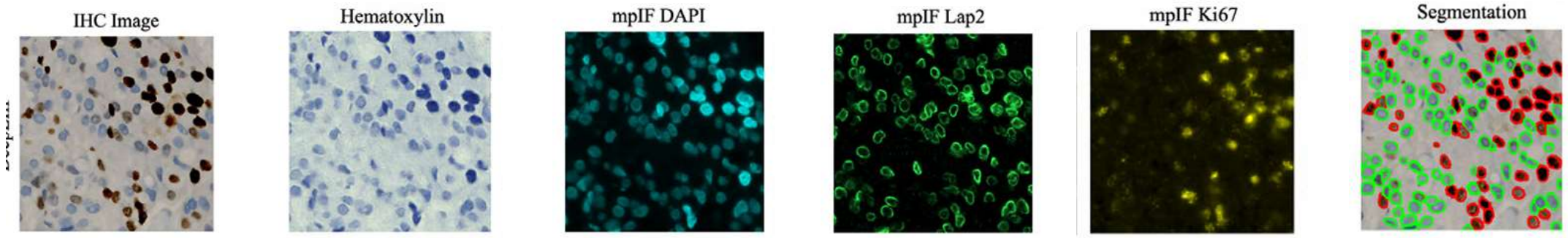
Work Done: They built an AI system that creates virtual stains from natural cell glow (autofluorescence). It made it easier to see important cancer markers like PD-L1, CD3, and CD8 without damaging real tissue samples.

How our project is Similar to them: In the 1st reference they used cancer markers like Ki67, CD3, and BCL2. We have also used markers like Hematoxylin, DAPI, LAP2, Ki67.

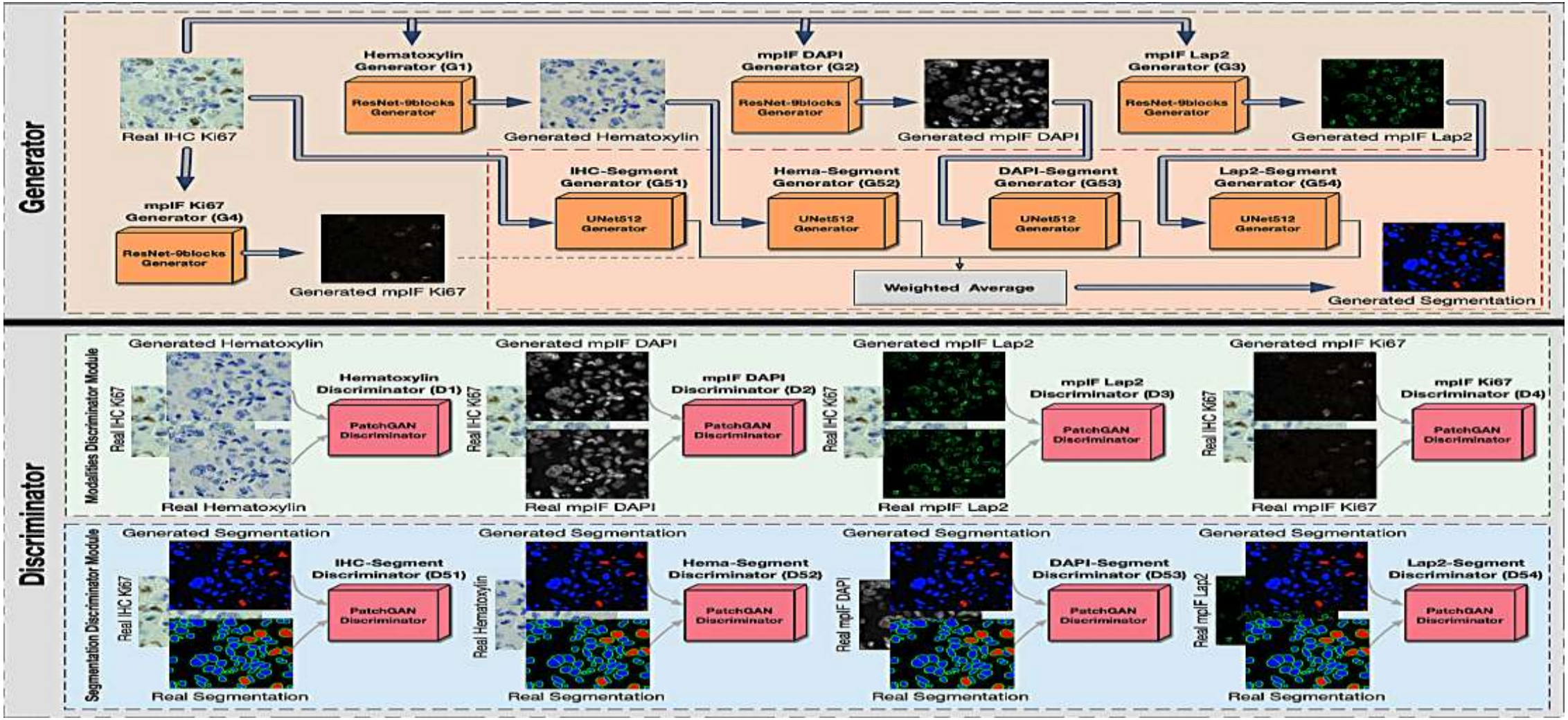
How our project is Different to them: We have used a number of models as discriminator like unet, unet++, ResNet, special block like CBAM(Convolutional Block Attention Module), SE(Squeeze-and-Excitation), SwingUNet, DuckNet, AttentionUNet which no one used earlier as a combination. Also we have used Weakly Supervised Learning(WSL) here which is a new concept in this field.

What Kind of Data Did We Use?

- The dataset consists of 1250 of IHC and immunofluorescence images, each split into six aligned modalities (IHC, Hematoxylin, DAPI, Lap2, Ki67, and segmentation mask).
- The segmentation task was framed as a binary classification problem (nuclei vs. background), while modality translation involved four target modalities.
- The dataset was divided into training (80%), validation (10%), and testing (10%) sets. For debugging and explainability analysis, we also used a reduced subset of the first 50 images.



Training Method



The regularization and optimization used

Regularization:

- Data augmentation (random flips, rotations, color jitter, etc.) was used to reduce overfitting and improve model generalization.
- Weight decay (L2 regularization) was applied during optimization to constrain parameter growth and stabilize training.
- Early stopping was monitored on validation loss to avoid overfitting by halting training once performance plateaued.

Optimization:

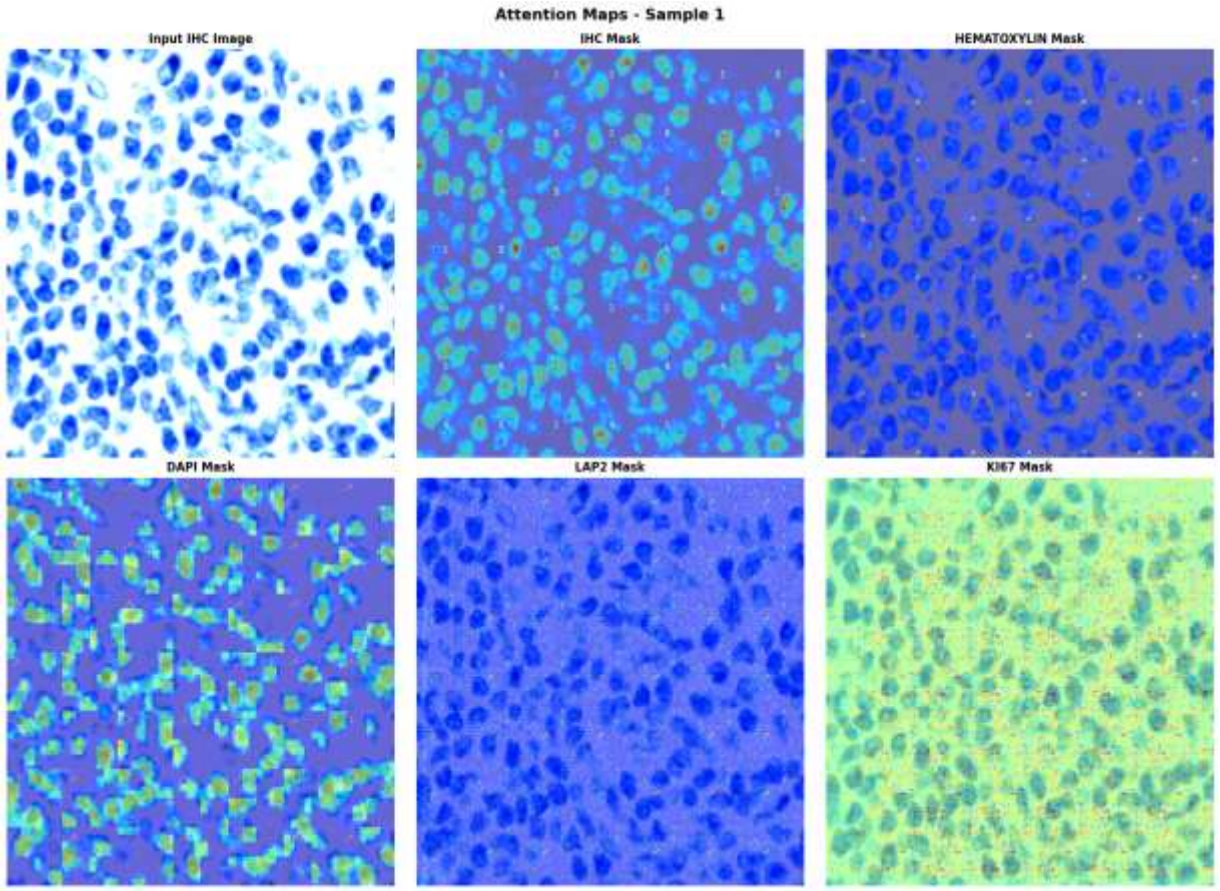
- The model was trained using the Adam optimizer, which adapts learning rates individually for each parameter and accelerates convergence in deep networks.
- The segmentation network used a hybrid loss function combining Dice loss, MSE, AJI and IoU, ensuring both overlap accuracy and pixel-wise correctness.

Results

Ablation table

Models	Dice	Parameters	Pixel Accuracy
Unet + UNet++ + WSL	0.6814	95M	95.54%
ResNet + SwinUNet +CBAM	0.5214	213M	93.45%
ResNet+AttentionUNet+SE	0.4557	78M	90.78%
ResNet + DuckNet + CBAM	0.4552	146M	88.87%

GradCam + Attention Map



Discussion on Results

- The combination of Dice + IoU + AJI + MSE losses ensures accurate boundary detection and robust performance in crowded cellular environments.
- Grad-CAM visualizations and attention maps indicate that The model focuses primarily on nuclei and biologically meaningful structures and attention is minimal in background regions, reducing false positives.
- In our best result we have used Unet as a generator and Unet++ as a discriminator. Use of WSL (Weakly Supervised Learning) make the model to achieve best result.
- Another best result structure was combination of ResNet and SwingUNet. Here we have used ResNet as generator and SwinUNet as discriminator. Use of CBAM have enhanced performance by adding spatial attention here.
- Next we got a good result by using ResNet as generator and AttentionUNet as discriminator. Use of SE have improved results by reweighting important feature channels, allowing the network to emphasize biologically relevant signals.